

WEEK - 04

Task 1: Network Scanning with Nmap

Objective: Learn how to scan a network and identify open ports.

Install Nmap and scan a local network.

Perform different types of scans:

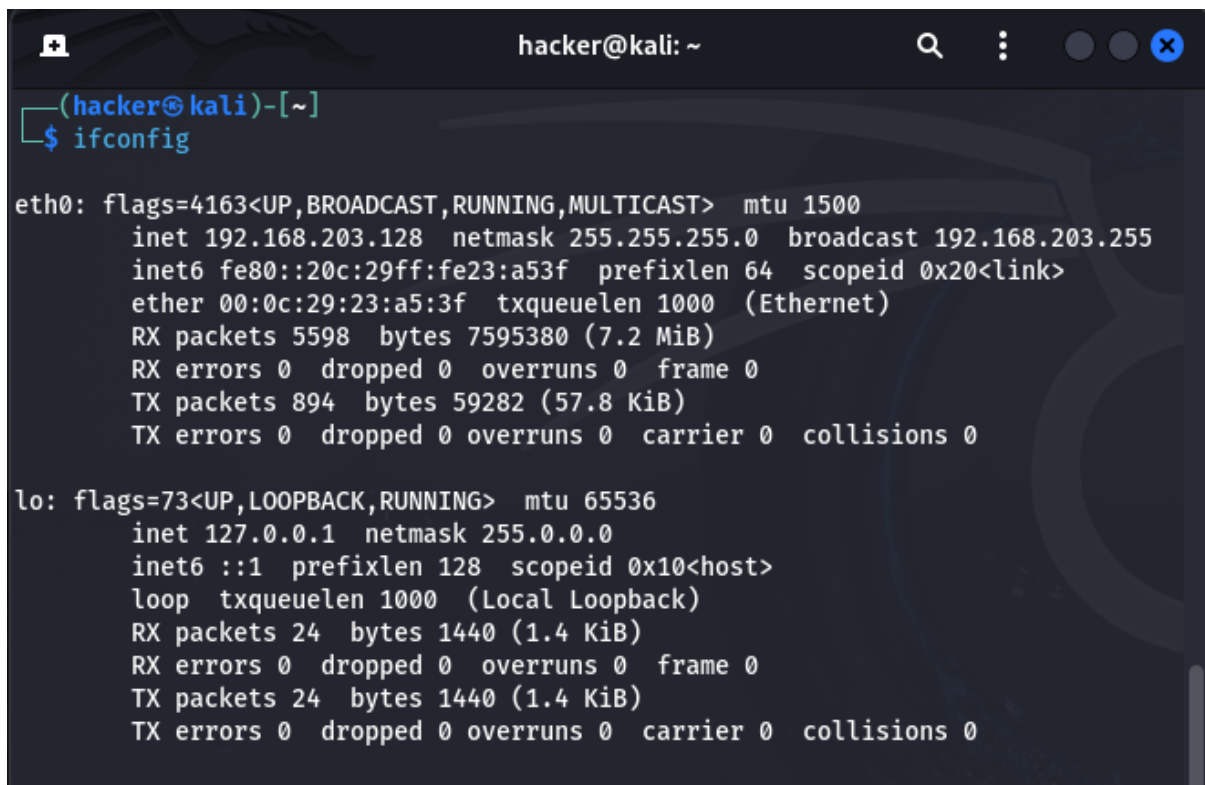
Basic scan: `nmap <IP>`

Service detection: `nmap -sV <IP>`

Aggressive scan: `nmap -A <IP>`

Submit a report on your findings (What services are running? Any vulnerabilities detected?) as a document, in WEE4 day 1

1.IP FINDING BY NMAP



```
hacker@kali: ~  
(hacker@kali)-[~]  
$ ifconfig  
  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.203.128 netmask 255.255.255.0 broadcast 192.168.203.255  
    inet6 fe80::20c:29ff:fe23:a53f prefixlen 64 scopeid 0x20<link>  
    ether 00:0c:29:23:a5:3f txqueuelen 1000 (Ethernet)  
    RX packets 5598 bytes 7595380 (7.2 MiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 894 bytes 59282 (57.8 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 24 bytes 1440 (1.4 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 24 bytes 1440 (1.4 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2.RUNNING PORT FINDING THROUGH A SPECIFIED IP

```
(hacker@kali)-[~]  
$ nmap -p- 192.168.203.128  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-27 22:41 EDT  
Nmap scan report for 192.168.203.128  
Host is up (0.000016s latency).  
Not shown: 65534 closed tcp ports (reset)  
PORT      STATE SERVICE  
1716/tcp  open  xmsg  
  
Nmap done: 1 IP address (1 host up) scanned in 2.71 seconds
```

3. SERVICE DETECTION THROUGH NMAP :

```
(hacker@kali)-[~]  
$ nmap -sV -p 1716 192.168.203.128  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-27 22:43 EDT  
Nmap scan report for 192.168.203.128  
Host is up (0.00015s latency).  
  
PORT      STATE SERVICE      VERSION  
1716/tcp  open  tcpwrapped  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 2.72 seconds
```

4.AGGRESIVE SCAN THROUGH THROUGH NMAP :

```
(hacker@kali)-[~]  
$ nmap -A -p 1716 192.168.203.128  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-27 22:44 EDT  
Nmap scan report for 192.168.203.128  
Host is up (0.00016s latency).  
  
PORT      STATE SERVICE      VERSION  
1716/tcp  open  tcpwrapped  
Warning: OSScan results may be unreliable because we could not find at least 1 o  
pen and 1 closed port  
Device type: general purpose  
Running: Linux 2.6.X|5.X  
OS CPE: cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:5 cpe:/o:linu  
x:linux_kernel:6  
OS details: Linux 2.6.32, Linux 5.0 - 6.2  
Network Distance: 0 hops  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 9.52 seconds
```

FINDINGS :

1. Port 1716 (TCP): The service is marked as tcpwrapped, which suggests the service is protected by access controls or encryption. Without credentials or additional probing, detailed information remains hidden.

2. Host Information:

- Device Type: General purpose system.
- Operating System: Detected as Linux with versions ranging from 2.6.32 to 6.2.
- Network Distance: 0 hops, indicating the target is on the same network segment.

3. OSScan Warnings: The warning about unreliable OS scanning suggests a lack of sufficient open and closed ports to improve detection accuracy. This is expected since only one port (1716) is open, and the others are closed.